SHAMROCK CAPITAL
Pricing the Masters of Taylor Swift

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SHAMROCK CAPITAL

- Music executive Braun has approached investment company Shamrock Capital with a proposal:
  - Acquiring the master recordings of Taylor Swift's first six albums.
- Reynolds, an analyst at Shamrock, is advising the partners at Shamrock regarding a bid to Braun;
- There is interest from other parties, including Taylor Swift;
- Speculation by insiders suggests that a winning bid could exceed $400m;
- Could Taylor Swift's master recordings be worth $400m+?

FORECASTED BASE-CASE

Exhibit 7

Shamrock Capital: Pricing the Masters of Taylor Swift

Estimated Number of Global Streams across All Platforms by Year for Swift Masters
(in millions)

Album	2018	2019	2020E	2021E	2022E	2023E	2024E	2025E	2026E
Taylor Swift	400	508	545	708	850	1,020	1,173	1,290	1,419
Fearless	864	1,357	1,528	1,986	2,383	2,860	3,289	3,618	3,980
Speak Now	565	888	1,033	1,343	1,612	1,934	2,224	2,447	2,692
Red	759	2,149	2,302	2,993	3,592	4,310	4,956	5,452	5,997
1989	1,116	1,382	3,277	4,260	5,112	6,135	7,055	7,760	8,536
Reputation	1,993	2,895	3,432	4,461	5,354	6,425	7,388	8,127	8,940
Total	5,697	9,181	12,117	15,752	18,903	22,683	26,085	28,694	31,563
Past growth-rate		61%	32%						
Forecasted growth-rate				30%	20%	20%	15%	10%	10%

Note: Taylor Swift removed her music from all streaming services between 2014 and 2017.
Source: Author estimates based on data reported by Spotify and analyst forecasts.

FORECASTED BASE-CASE Income Statements (EFP)

Exhibit 10

Shamrock Capital: Pricing the Masters of Taylor Swift
Financial Forecast for Swift Catalog (in millions)

Assumptions

	2020E	2021E	2022E	2023E	2024E	2025E	2026E
Total streams	12,117	15,752	18,903	22,683	26,085	28,694	31,563
Master royalty rate per stream	$0.0038	$0.0038	$0.0039	$0.0039	$0.0040	$0.0040	$0.0041
Revenue-sharing rate with artist	22.0%	22.0%	22.0%	22.0%	22.0%	22.0%	22.0%
Revenue-sharing rate with Braun		3.0%	3.0%	3.0%	3.0%	3.0%	3.0%
Third-party operating expense %		36.0%	36.0%	36.0%	36.0%	36.0%	36.0%
Receivable and accrued royalty days		60.0	60.0	60.0	60.0	60.0	60.0

Income Statement

	2020E	2021E	2022E	2023E	2024E	2025E	2026E
Revenue	48.2	70.8	83.8	99.7	114.6	126.6	140.1
Royalty sharing with artist	10.6	15.6	18.4	21.9	25.2	27.8	30.8
Royalty sharing with Braun		2.1	2.5	3.0	3.4	3.8	4.2
Third-party operating expenses		25.5	30.2	35.9	41.2	45.6	50.4
Operating profit		27.6	32.7	38.9	44.7	49.4	54.7
Shamrock Receivables		4.5	5.4	6.4	7.3	8.1	9.0

Source: Author analysis.

BASE-CASE DCF VALUATION

- Explicit forecasting period: 2021-2026;

- Main assumptions:
  - ✓Swift's streams will grow from 30% in 2021 to 10% in 2026;
  - ✓Swift's physical copies will remain constant;
  - ✓The royalty rate will grow by 1.5% per year.

- Workings:

  Shamrock's Receivables: (60/365) Operating Profit

BASE-CASE DCF VALUATION

Exhibit 11

Shamrock Capital: Pricing the Masters of Taylor Swift

Valuation Model for Swift Catalog. Base-Case Scenario with No Rerecording (in millions)

Assumptions	
Cost of capital	8.0%
Steady-state growth rate	2.0%
Tax rate	21%

	2020E	2021E	2022E	2023E	2024E	2025E	2026E
Operating profit		27.6	32.7	38.9	44.7	49.4	54.7
Receivables	-	4.5	5.4	6.4	7.3	8.1	9.0
Annual Cash Flow and Terminal Value							
Net operating profit after tax (NOPAT)		21.8	25.8	30.7	35.3	39.0	43.2
Change in receivables		4.5	0.8	1.0	1.0	0.8	0.9
Free cash flow		17.3	25.0	29.7	34.3	38.2	42.3
						Steady-state cash flow	43.9
						Terminal value	731

Discounted Cash Flow	
Present value of forecast horizon	139
Present value of terminal value	461
Total value to Shamrock	600

Source: Author analysis.

BASE-CASE DCF VALUATION
Cost of Capital Estimation

Exhibit 8

Shamrock Capital: Pricing the Masters of Taylor Swift
Current Financial and Capital Market Data for Industry Comparable Companies

Financial Data

	Revenue (in millions of US	EBIT (in millions of US dollars)	Total Debt (in millions of US dollars)	Share Price	Shares Outstand. (in millions)	Bond Rating	Beta
Spotify	6,764	-73	622	149	184	NA	1.25
Netflix	20,156	2,604	14,759	329	439	A	1.25
Disney	69,570	11,851	46,986	144	1,781	AA	1.20
Warner Music	4,457	356	2,974	32	157	BB+	1.50
Sirius XM	7,794	1,731	8,346	7	4,412	BB+	1.40

Business Description

Spotify	Online music streaming services
Netflix	Online video entertainment
Disney	Diversified provider of entertainment services
Warner Music	Music entertainment
Sirius XM	Online entertainment services

Sources: Yahoo! Finance and author estimates.

BASE-CASE DCF VALUATION
Cost of Capital Estimation

- For each company, derive the:
  - ✓ cost of equity capital (CAPM);
  - ✓ cost of the firm capital (WACC formula).
- Derive the average WACC;
- Use it as the appropriate cost of capital.

Exhibit 9

Shamrock Capital: Pricing the Masters of Taylor Swift
Capital Markets Data, September 2020

Prevailing Yields	
US government yields:	
90-day Treasury	0.12%
20-year Treasury	1.21%
Corporate bond yields by credit rating:	
AAA	1.76%
AA	2.25%
A	2.79%
BBB	3.57%
BB	5.29%
B	6.91%
Historical Returns	
Average annual US returns (1955 to 2019)	
Large-cap stocks (S&P 500)	11.8%
Long-term government bonds	6.1%
Short-term government bills	4.4%

https://home.treasury.gov/resource-center/data-chart-center/interest-rates/TextView?type=daily_treasury_yield_curve&field_tdr_date_value=2020 (accessed Jul. 13, 2023); Ibbotson Stocks, Bonds, Bills, and Inflation Data set; and author estimates.

BASE-CASE DCF VALUATION
Cost of Capital Estimation

Cost of Capital Estimates

Risk-free rate	1.2%
Market risk premium	5.7%
Tax rate	21%

	D/V	Kd	Bond rating	Ke	Beta	WACC
Spotify	2%	3.57%	NA	8.3%	1.25	8.2%
Netflix	9%	2.79%	A	8.3%	1.25	7.8%
Disney	15%	2.25%	AA	8.1%	1.20	7.1%
Warner Music	37%	5.29%	BB+	9.8%	1.50	7.7%
Sirius XM	21%	5.29%	BB+	9.2%	1.40	8.1%
Average						7.8%

Note: WACC = weighted-average cost of capital; D/V = debt / firm value; Kd = cost of debt; Ke = cost of equity.
Source: Author analysis.

BASE-CASE DCF VALUATION
Continuing Value Estimation

- In the first year following the end of the EFP, 2027, the steady state FCF is estimated as:

$$FCF_{2027} = NOPAT_{2026}*(1+g) – g*BVNOA_{2026}$$

- It follows that:

$$FCF_{2027} = \$43.2m*(1+2\%) – 2\%*\$9m = \$43.9m$$

$$CV_{2027} = FCF_{2027}/(WACC-g) = \$43.9m/(8\%-2\%) = \$731.7m$$

BASE-CASE DCF VALUATION

Exhibit 11

Shamrock Capital: Pricing the Masters of Taylor Swift

Valuation Model for Swift Catalog. Base-Case Scenario with No Rerecording (in millions)

Assumptions							
Cost of capital	8.0%						
Steady-state growth rate	2.0%						
Tax rate	21%						
	2020E	**2021E**	**2022E**	**2023E**	**2024E**	**2025E**	**2026E**
Operating profit		27.6	32.7	38.9	44.7	49.4	54.7
Receivables	-	4.5	5.4	6.4	7.3	8.1	9.0
Annual Cash Flow and Terminal Value							
Net operating profit after tax (NOPAT)		21.8	25.8	30.7	35.3	39.0	43.2
Change in receivables		4.5	0.8	1.0	1.0	0.8	0.9
Free cash flow		**17.3**	**25.0**	**29.7**	**34.3**	**38.2**	**42.3**
Discounted Cash Flow							
Present value of forecast horizon	139				Steady-state cash flow		43.9
Present value of terminal value	461				Terminal value		731
Total value to Shamrock	**600**						

Source: Author analysis.

BASE-CASE DCF VALUATION

Forecast for Swift	2020E	**2021E**	**2022E**	**2023E**	**2024E**	**2025E**	**2026E**
Royalty income	10.6	15.6	18.4	21.9	25.2	27.8	30.8
Receivable days	60	60	60	60	60	60	60
Receivables	1.7	2.6	3.0	3.6	4.1	4.6	5.1
Cash Flow to Swift							
NOPAT		12.3	14.6	17.3	19.9	22.0	24.3
Free cash flow		11.5	14.1	16.7	19.4	21.6	23.9
						Steady-state cash flow	24.7
						Terminal value	412
Value to Swift							
Discounted cash flow							
Present value of forecast horizon	80						
Present value of terminal value	260						
Total value to Swift	**340**						

BASE-CASE DCF VALUATION

- How realistic is this valuation?
  - ✓Forecast horizon cash flow estimates?
  - ✓Cost of capital including choices of peers?
  - ✓Long-term growth rate?
- Would you expect Taylor Swift to re-record?

BASE-CASE DCF VALUATION

- How realistic is this valuation?
  - ✓ Forecast horizon cash flow estimates?
    Too aggressive?
  - ✓ Cost of capital including choices of peers?
    Err…
  - ✓ Long-term growth rate?
    About right?
- Would you expect Taylor Swift to re-record?
  Let us check…

WILL TAYLOR SWIFT RE-RECORD?

- Taylor Swift:

  - ✓ believes that artists deserve to own the art they created;
  - ✓ removed her entire discography from Spotify from 2014 to late 2017 to protest its payment structure to artists;
  - ✓ withheld her music in another protest from Apple for a few months in 2015 and won her battle.

DCF VALUATION
Taylor Swift Re-records Scenario

Shamrock Capital: Pricing the Masters of Taylor Swift

Estimated Number of Global Original Masters Song Streams (in millions) by Year across All Platforms. Scenario: Swift Rerecords

Erosion Factor Estimate of Original Masters Volume Relative to Release Year of Rerecorded Album

Year 0	Year +1	Year +2 and beyond
-50%	-65%	-80%

Old Master Recording

	Estimated rerelease year	2020E	2021E	2022E	2023E	2024E	2025E	2026E
Taylor Swift	2024	545	708	850	1,020	586	451	284
Fearless	2021	1,528	993	834	572	658	724	796
Speak Now	2023	1,033	1,343	1,612	967	779	489	538
Red	2022	2,302	2,993	1,796	1,508	991	1,090	1,199
1989	2024	3,277	4,260	5,112	6,135	3,527	2,716	1,707
reputation	2025	3,432	4,461	5,354	6,425	7,388	4,064	3,129
Total streams		12,117	14,759	15,558	16,626	13,930	9,534	7,654

New Master Recording

	2020E	2021E	2022E	2023E	2024E	2025E	2026E
Taylor Swift	-	-	-	-	586	838	1,135
Fearless	-	993	1,549	2,288	2,631	2,894	3,184
Speak Now	-	-	-	967	1,446	1,958	2,153
Red	-	-	1,796	2,801	3,965	4,362	4,798
1989	-	-	-	-	3,527	5,044	6,829
reputation	-	-	-	-	-	4,064	5,811
Total streams	-	993	3,345	6,057	12,156	19,160	23,910

Source: Author analysis.

DCF VALUATION
Taylor Swift Re-records Scenario

Shamrock Capital: Pricing the Masters of Taylor Swift

Valuation Model. Scenario: Swift Rerecords According to Case Exhibit 12 (in millions)

Assumptions

Cost of capital	8.0%	Steady-state growth rates	
Tax rate	21%	Old masters	2.0%
		New masters	2.0%

	2021E	2022E	2023E	2024E	2025E	2026E
Total streams of old masters	14,759	15,558	16,626	13,930	9,534	7,654
Total streams of new masters	993	3,345	6,057	12,156	19,160	23,910
Total streams	15,752	18,903	22,683	26,085	28,694	31,563

DCF VALUATION
Taylor Swift Re-records Scenario

Forecast for Shamrock	2020E	2021E	2022E	2023E	2024E	2025E	2026E
Revenue		67.0	70.9	76.0	66.3	49.4	42.2
Royalty sharing with artist		14.7	15.6	16.7	14.6	10.9	9.3
Royalty sharing with Braun		2.0	2.1	2.3	2.0	1.5	1.3
Third-party operating expenses		24.1	25.5	27.4	23.9	17.8	15.2
Operating profit		26.1	27.7	29.6	25.8	19.3	16.5
Receivables		4.3	4.5	4.9	4.2	3.2	2.7
CashFlow to Shamrock							
Net operating profit after tax (NOPAT)		20.6	21.9	23.4	20.4	15.2	13.0
Free cash flow		16.3	21.6	23.1	21.0	16.3	13.5
						Steady-state cash flow	13.2
						Terminal value	220
Value to Shamrock							
Present value of forecast horizon	87						
Present value of terminal value	139						
Total value to Shamrock	**226**						

DCF VALUATION
Taylor Swift Re-records Scenario

Forecast for Swift	2020E	**2021E**	**2022E**	**2023E**	**2024E**	**2025E**	**2026E**
Royalty income on old masters		14.7	15.6	16.7	14.6	10.9	9.3
Receivables on old masters	1.7	2.4	2.6	2.7	2.4	1.8	1.5
Royalty income on new masters		3.8	12.9	23.7	48.3	77.3	97.9
Cost of rerecording		20.0	20.0	20.0	40.0	20.0	
Third-party operating expenses		1.4	4.6	8.5	17.4	27.8	35.2
Receivables on new masters		0.6	2.1	3.9	7.9	12.7	16.1
Cash Flow to Swift							
Cash flow from old masters		11.0	12.2	13.0	11.9	9.2	7.6
Cash flow from new masters		-14.5	-10.8	-5.6	-11.2	18.5	46.1
Total cash flow		-3.6	1.4	7.4	0.6	27.7	53.7

Value to Swift			
Discounted cash flow			
Present value of forecast horizon	57	Terminal value old masters	124
Present value of terminal value (both)	605	Terminal value new masters	836
Total value to Swift	**662**		

Source: Author analysis.

DCF VALUATION
Valuation Model Summary

Shamrock Capital: Pricing the Masters of Taylor Swift
Valuation Model Summary (in millions)

Scenario: Base Case of No Rerecording

Value to Braun	47.0
Value to Swift	339.7
Value to Shamrock	599.6
Total	986.3

Scenario: Swift Re-records According to Case Exhibit 12

Value to Braun	17.4
Value to Swift	662.0
Value to Shamrock	225.9
Total	905.3
Deadweight Loss	81.0

Expected Value

Probability of Re-recording (according to case text) at 60%

Value to Braun	29.2
Value to Swift	533.1
Value to Shamrock	375.4

Source: Author analysis.

DCF VALUATION
Taylor Swift Re-records Scenario

- Taylor Swift is strictly better off re-recording: $662m vs $337.9m;

- Could re-recording jeopardize releasing new albums?

  Unlikely…

- Shamrock Capital should hence bid no more than $225.9m

  Or abstain from bidding…

DCF VALUATION
Epilogue

- Shamrock acquired Swift's master recordings for $300m in October 2020;

- Taylor Swift started re-recording in April 2021.